

AE 8883: Spacecraft Engineering I

Printed Circuit Boards (PCBs)

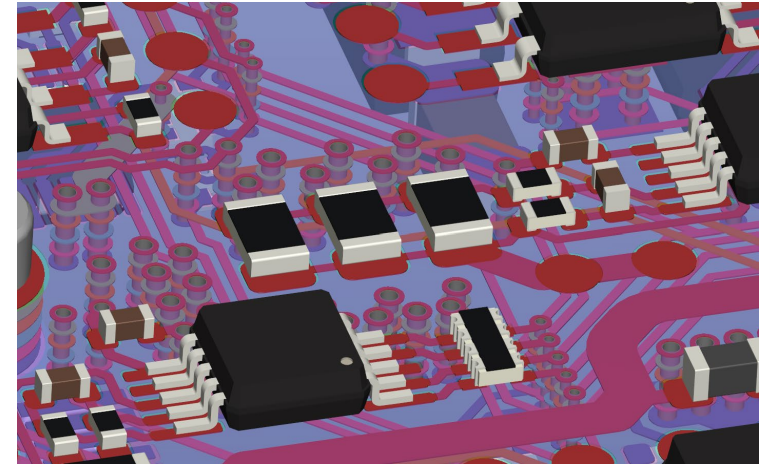
September 28, 2022



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Outline of today's seminar

- About PCBs
 - Structure and Materials
 - Layers
- PCB Manufacture
 - Popular Fabs
 - Fabs at GT
 - Testing and Validation
- PCB Design
 - Manufacturer Capabilities
 - Design Considerations

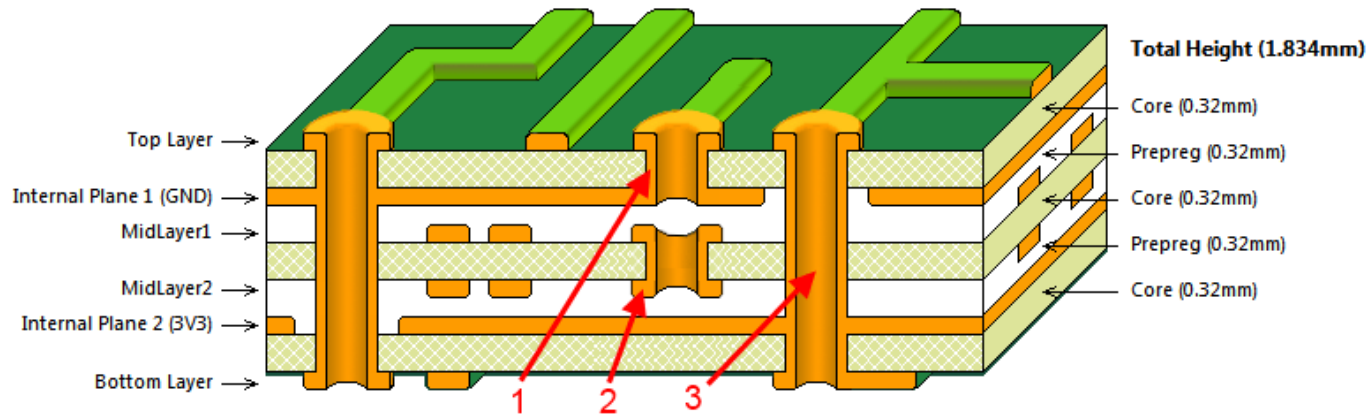




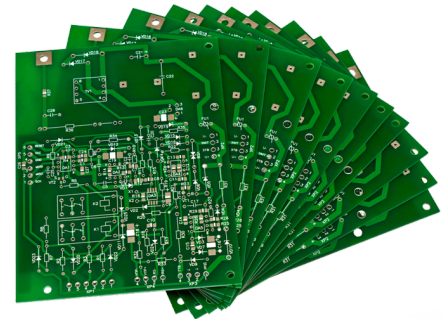
About PCBs

About PCBs

- Board consisting of fused, alternating layers of an insulating plastic substrate and circuitry foil made of conductive material (usually copper)
- Designed with attachment points for ICs and other circuit components to complete the circuit



https://www.pcbgogo.com/Article/An_Introduction_to_Multilayer_PCBs.html





Plastic Substrates

FR-4

- Standard fiberglass composite
- Aluminum-backed

- Useful for heatsinking purposes

Rogers

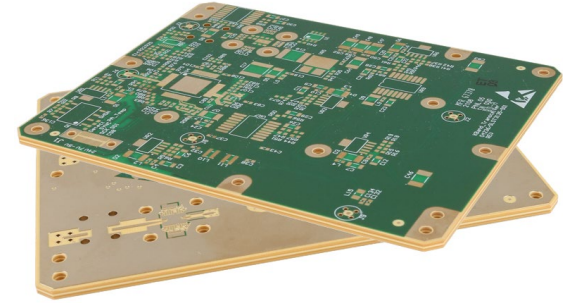
- fiberglass with ceramic laminate
used for high-frequency circuits

HDI (High Density Interconnect)

- Allows for buried and blind vias

PI (Polyimide) Film

- Used for flexible PCBs



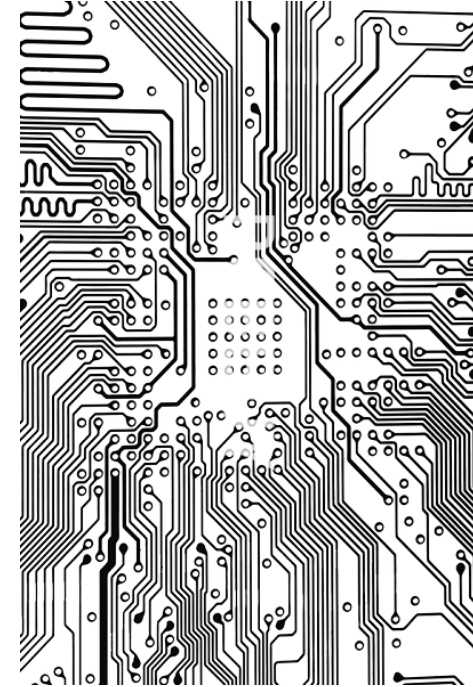
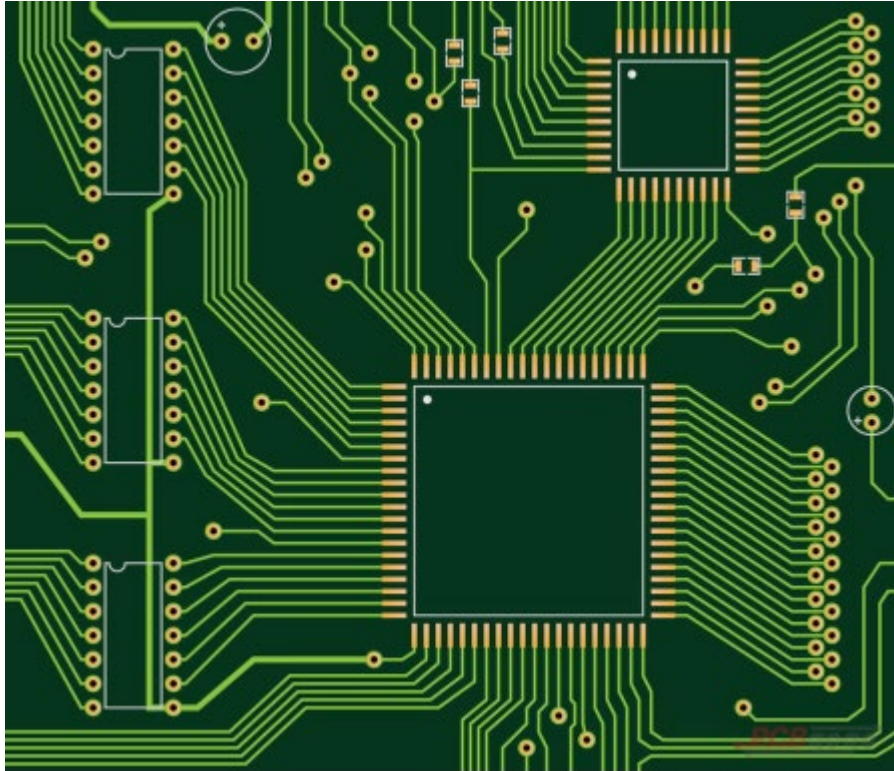
Plastic Substrates

Choose a substrate depending on your required:

- ⦿ T_g = glass transition temperature
- ⦿ D_k = dielectric constant
- ⦿ K = Thermal Conductivity
- ⦿ CTE = Thermal Coefficient of Expansion
- ⦿ (and other factors)

Traces

Lines of conductive material that carry signals around a PCB

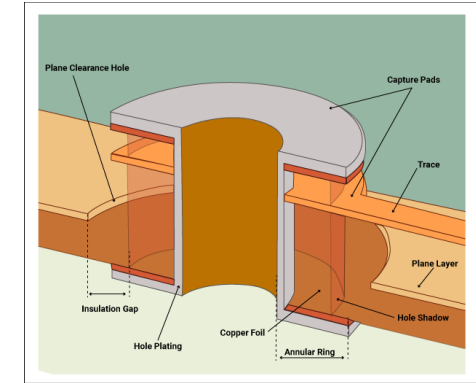


Vias

Holes lined with conductive metal that allow for connections between different layers of the PCB

- Essentially giant metal cylinders
- Blind and buried vias are possible, but dramatically increase costs

Note: Unlike vias, **holes** are not lined with conductive material, they are just drilled into the PCB.



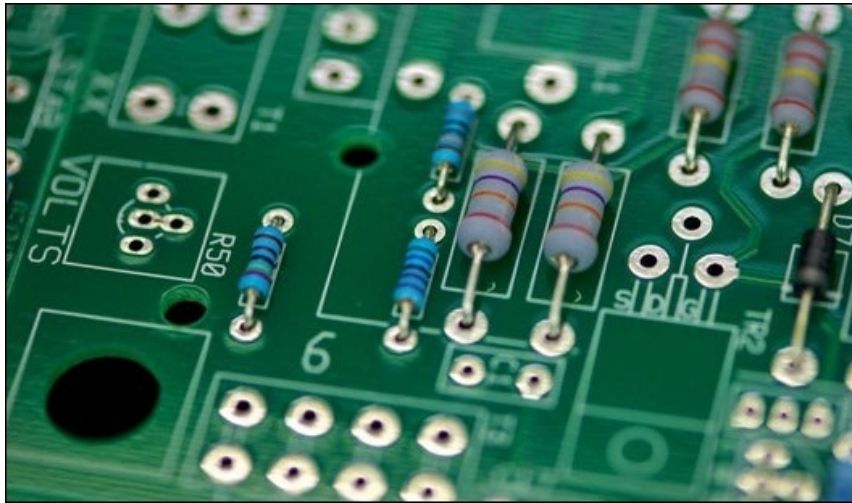
<https://www.protoexpress.com/blog/what-is-pad-pcb-design-development/>

Type	Cost		Duration	
	Standard	Standard	+\$	++
Through Via	Standard	Standard	+	+
Tented Via	Standard	Standard	+	+
Blind Via	Standard	Standard	+	+
Buried Via	Standard	Standard	+	+

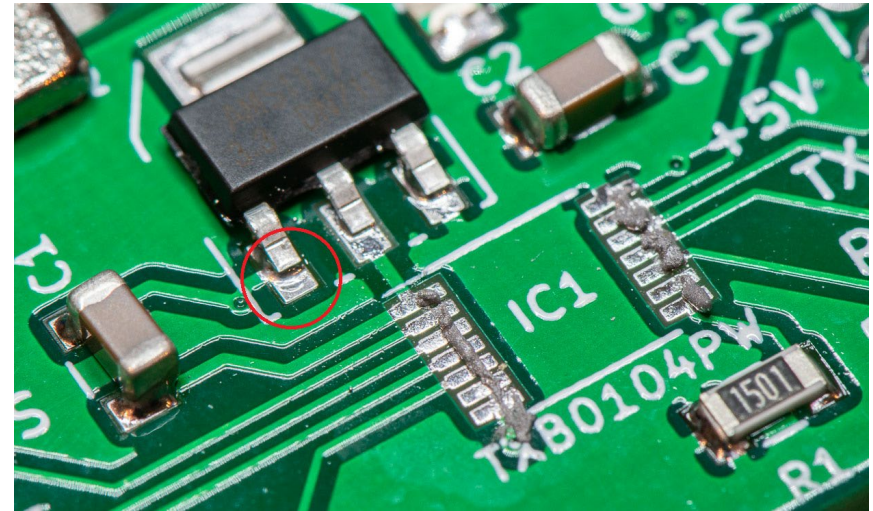
<https://www.tempoautomation.com/blog/via-in-pad-guidelines-choosing-the-best-routing-for-your-smbs-tempo-automation/>

Vias and Pads for Mounting Components

Through-Hole Vias

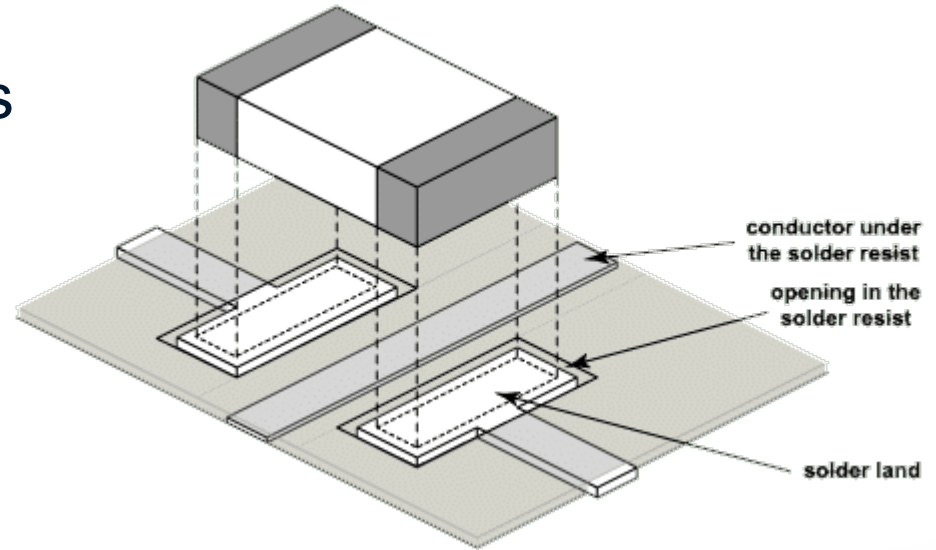


SMD Pads



Solder Mask

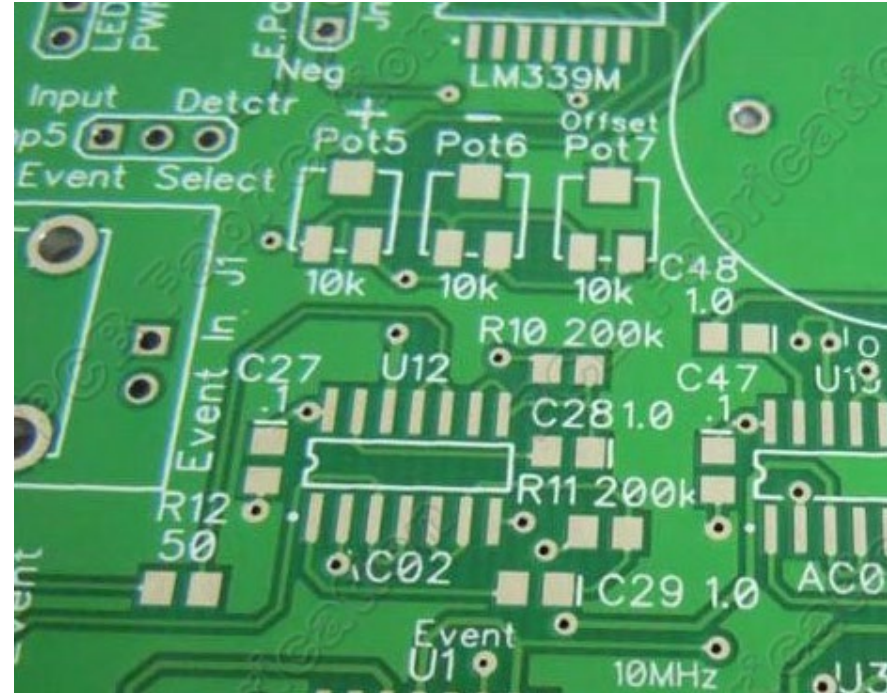
- Mask that prevents solder from wicking onto certain parts of the PCB
- Should leave pads and through-holes exposed for you to solder components onto
- Should cover other areas



<https://passive-components.eu/pcb-board-design-and-handling/>

Silkscreen

- Used for helpful labels, funny pictures, and everything in-between
- Component names and reference designators are helpful to see if manufacturing by hand



Conformal Coating Considerations

- Thin polymeric film applied to PCB that protects it from environmental hazards
- Can insulate board and prevent electric arcing from static or high voltages
- Flight PCBs should use low outgassing conformal coatings
 - Parylene (offers void-free conformal coating due to the required deposition process)
 - Arathene 5750 (faster processing time)



PCB Manufacture

Manufacturing Process



PCB starts as a fiberglass sheet about 16" x 20" x .062"



Thin copper sheets are added to both sides



Etchant-resistive wire pattern printed on one or both sides



All copper removed with etchant except for wires and pads



Holes drilled for leads and vias, and plated with metal

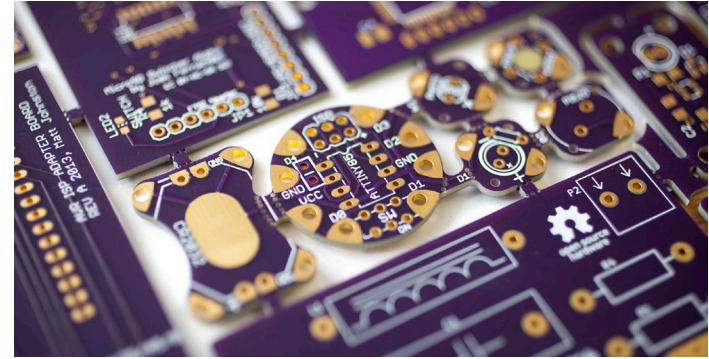


Soldermask applied to keep solder only where needed

1. Starting substrate is referred to as the **core** of the PCB
2. Copper is attached to both sides of the core, and the trace pattern is printed onto the copper using an etchant-resistive material
3. Chemicals etch the surrounding copper away, leaving just the traces
4. Vias and holes are drilled
5. If multi-layered, additional substrate layers called **prepreg** are added, followed by more copper etching
6. Soldermask and silkscreen are printed on the top and bottom

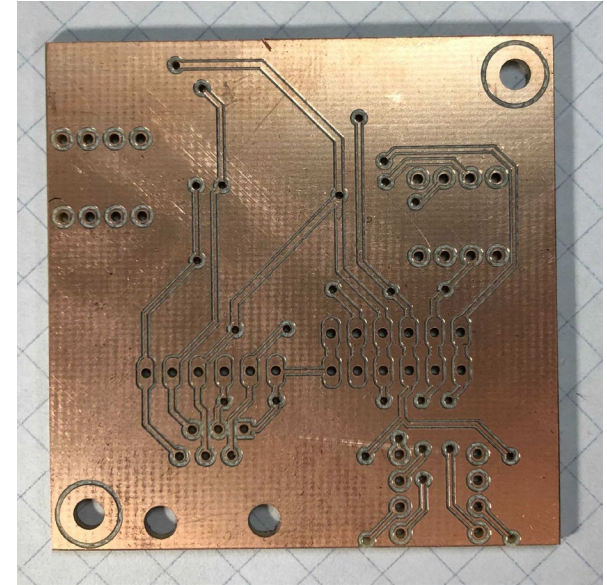
Popular PCB Fabricators

- OSH Park
 - Fast, US located, expensive
- JLC
 - Fast, low cost, CN based
- Sierra Circuits
 - More complex boards, US located, very expensive
- Sunstone Circuits
 - More complex boards, US located, expensive



PCB Fabrication at GT

- For simple boards or size mockups, have it made on campus!
- Limitations:
 - Two layer only
 - No solder mask or silkscreen
 - Large features only
 - Must solder or rivet vias
- ME Electronics Lab
 - All done for you, for free!
<https://www.me.gatech.edu/electronics-lab>
- The IDC (Hive)
 - DIY. Get yourself trained:
<http://wiki.hive.ece.gatech.edu/doku.php?id=pcb-training>



Acceptance Testing

- ⦿ Upon receiving the PCB, check for continuity between appropriate signal lines, and keep an eye out for broken/abnormal traces
 - If you receive more copies of a board than you ordered, chances are one of them contained a manufacturing defect
- ⦿ After soldering and assembly, verify the functionality of the circuit given both reasonable and unreasonable inputs
 - Check that your circuitry can handle abnormal inputs and outputs gracefully, within your defined requirements
 - Functionality of subcircuits can be checked by designing and manufacturing breakout boards containing just the relevant circuitry

Environmental Testing

You'll also want to verify the assembled PCB can withstand the hazards expected during flight and in space

- ⦿ Temperature Testing

- Heat cycle the assembled PCB and bring it to the expected temperature extremes

- ⦿ Vibration Testing

- Solder joints are brittle and can break if the PCB is flexed

- ⦿ Vacuum Testing

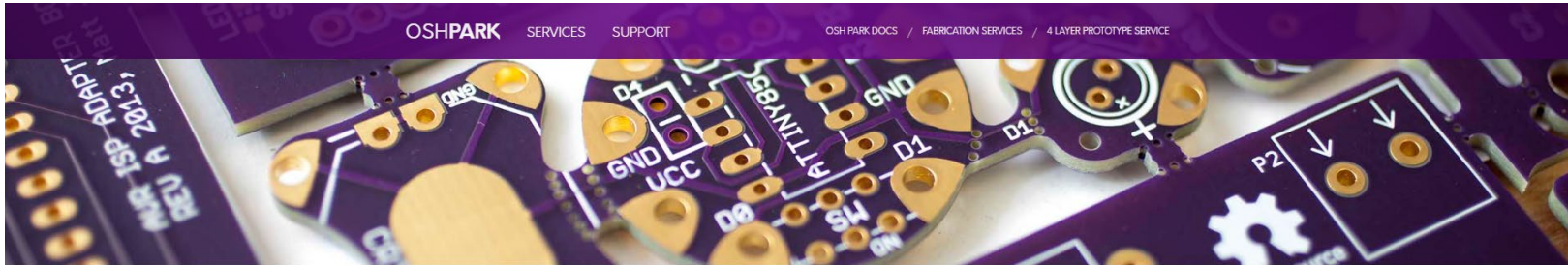
- ⦿ Radiation Testing



PCB Design

PCB Manufacturer Capabilities

Check your manufacturer's website for important information regarding what their fabrication processes support



4 Layer Prototype Service

Pricing

\$10 per square inch, which includes three copies of your design.

For example, a 2 square inch board would cost \$20 and you'd get three copies of your board. You can order as many copies as you want, as long as they're in multiples of three.

Turn Times

Orders will be shipped within 14 calendar days of ordering.

You can get a quote, approve a design, and pay for an order at [OSH Park](#).

We do not offer Super Swift Service for 4 layer boards.

<https://docs.oshpark.com/services/four-layer/>



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Design Laboratory

PCB Manufacturer Capabilities

Stackup

- Outlines layer order and thickness
- Used during trace width calculations

Stackup

Thickness	Layer	Tolerance	dk
1mil (0.0254mm)	 silkscreen	+/-0.2mil (0.00508mm)	
1 mil (0.0254mm)	 solder resist	+/-0.2mil (0.0051mm)	
1.7 mil (0.0432mm)	 1 oz copper clad+plated		
7.96 mil (0.2021mm)	 FR408HR 2113 prepreg	+/-0.796mil (0.0202mm)	3.61 at 1GHz
0.68 mil (0.0175mm)	 0.5 oz copper clad		
39 mil (0.9906mm)	 FR408HR core	+/-3.9mil (9.906mm)	
0.68 mil (0.0175mm)	 0.5 oz copper clad		
7.96 mil (0.2021mm)	 FR408HR 2113 prepreg	+/-0.796mil (0.0202mm)	3.61 at 1GHz
1.7 mil (0.0432mm)	 1 oz copper clad+plated		
1 mil (0.0254mm)	 solder resist	+/-0.2mil (0.0051mm)	
1mil (0.0254mm)	 silkscreen	+/-0.2mil (0.00508mm)	

<https://docs.oshpark.com/services/four-layer/>

PCB Manufacturer Capabilities

Material Specs

- Datasheets on materials used
- Thickness and dielectric are good to know for high frequency circuits

Material Specs

Spec	Value	
Substrate	190Tg FR408-HR	ISOLA FR408-HR Datasheet PDF
Board Thickness	63mil (1.6mm) nominal	
Dielectric	3.61 at 1GHz	
Soldermask Color	Purple	Mask Datasheet
Minimum soldermask web	4 mil (0.1016mm)	
Maximum soldermask alignment	3mil (0.0762mm)	Covers retraction, expansion, and shift
Silkscreen minimum line width	5 mil (0.127mm) (recommended minimum) 3 mil (0.0762mm) (short lines, text, graphics)	Silkscreen Datasheet
Maximum board size	16in (406.4mm) by 22in (558.8mm)	
Minimum board size	0.25in (6.35mm) by 0.25in (6.35mm)	

<https://docs.oshpark.com/services/four-layer/>

PCB Manufacturer Capabilities

Copper Specs

- Weight (thickness) is used for trace width calculations
- Lists minimum manufacturable trace spacings, trace widths, and via ring sizes

Copper Specifications

Spec	Value
Copper Layers	4
Copper Weight	1oz outer 1/2oz inner
Trace Spacing	5mil (0.127mm)
Trace Width	5mil (0.127mm)
Annular Ring	4mil (0.1016mm)
Board Edge Keepout	15mil (0.381) from nominal board edge
Via Plating Thickness	1mil (0.0254mm)

<https://docs.oshpark.com/services/four-layer/>

PCB Manufacturer Capabilities

Drill Specs

- Has information on the minimum sizes you can drill holes and vias

Drill Specifications

Spec	Value	
Minimum Annular Ring	4mil (0.1016mm)	
Minimum Drill Size	10mil (0.254mm)	
Minimum Slot Size	20mil (0.508mm) (drill slot only)	Additional information on slots
Drill Size tolerance	Max: +/- 2.5mil (0.0635mm) Typical: +/- 1.0mil (0.0254)	
Drill Positional Tolerance	Max: 2mil (0.0508mm) Typical: <1mil (0.0254mm)	
Via Tenting	Yes (filled hole and flat surface not guaranteed)	
Buried Via	No	
Blind Via	No	
Overlapping drills	Allowed, but not guaranteed. May result in missing or slotted holes. 5 mil (0.127mm) clearance is recommended between holes.	
Castellations	Allowed, but not guaranteed	Details and recommendations
Maximum Drill Size	None	Drill sizes above 250mil (6.35mm) will I

<https://docs.oshpark.com/services/four-layer/>

DRC (Design Rules Check)

- Most PCB fabs will provide a DRC file that you can use in conjunction with your EDA software to automatically enforce and verify that your design is manufacturable
- Import this DRC before doing any board design

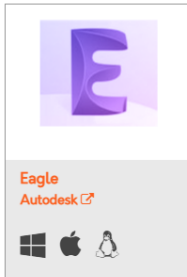
OSHPARK

SERVICES

SUPPORT

OSH PARK DOCS / DESIGN TOOL HELP / EAGLE / DESIGN RULES FILES

Design Rules Files.



When laying out your design, it's advised to load our Design Rules, so Eagle will generate warnings and ground planes according to our manufacturing specs.

Downloading CAM files

We have published some scripts to automatically download, install, and update our files for Eagle. You'll find those here: [OSH Park Eagle Tools](#)

If you prefer to install them manually, you can find our basic DRU files here:

[oshpark 2layer.dru](#)

[oshpark 4layer.dru](#)

[oshpark 6layer.dru](#)

After downloading the file, you can load it under **Tools > DRC > Load**, and then selecting the appropriate file.

User Adjustable Settings

In addition to manufacturing requirements, the DRC settings also contain a few user-

<https://docs.oshpark.com/design-tools/eagle/design-rules-files/>

Table of Contents

- [Downloading CAM files](#)
- [User Adjustable Settings](#)



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Trace Width and Design

The recommended width of a trace is calculated based on various factors:

- Trace thickness
- Trace width
- Amount of current the trace carries
- Trace Length
- Ambient Temperature

Note that most calculators assume that the external layer is in contact with air... not true in space!

Trace width calculator tries to control the temperature rise of the traces; it makes the internal traces wider. In vacuum, or in a potted assembly, you should use the *internal* layer guidelines even for the *external* layers.

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 PCB's Made in the USA Home / PCB Trace Width Calculator

PCB Trace Width Calculator

Printed Circuit Board Width Tool

This Javascript web calculator calculates the trace width for printed circuit board conductors for a given current using formulas from IPC-2221 (formerly IPC-D-275).

Inputs:

Current	10	Amps
Thickness	2	oz/ft ² ▾

Optional Inputs:

Temperature Rise	10	Deg C ▾
Ambient Temperature	25	Deg C ▾
Trace Length	1	Inch ▾

Results for Internal Layers:

Required Trace Width	368	mil ▾
Resistance	0.000685	Ohms
Voltage Drop	0.00685	Volts
Power Loss	0.0685	Watts

Results for External Layers in Air:

Required Trace Width	142	mil ▾
Resistance	0.00178	Ohms
Voltage Drop	0.0178	Volts
Power Loss	0.178	Watts

<https://www.4pcb.com/trace-width-calculator.html>

Heat Dissipation

- Substrates have some degree of heat sink capability, such as FR-4
 - Metal substrates exist too, but \$\$\$
- Exposed pads (usually GND, but not electrically connected!) underneath hot components can act as thermal paths that conduct heat away
- Via stitching can also help distribute heat
 - PCBs are generally more conductive across layers than they are through the board
 - You can use vias to bring heat from one side of the board to the other

Parasitics

- Unintentional resistances, capacitances, and inductances caused by your layout decisions
- Are always present ☹ but can be reduced
- Can be simulated in detail with a 3D field solver
- Can be simulated more simply using other methods (I.e. add RLC components into your circuit)

Parasitic Resistances

- Everything in every circuit has some resistance (even our conductive copper traces)
- Mostly applies to signals in series with one another

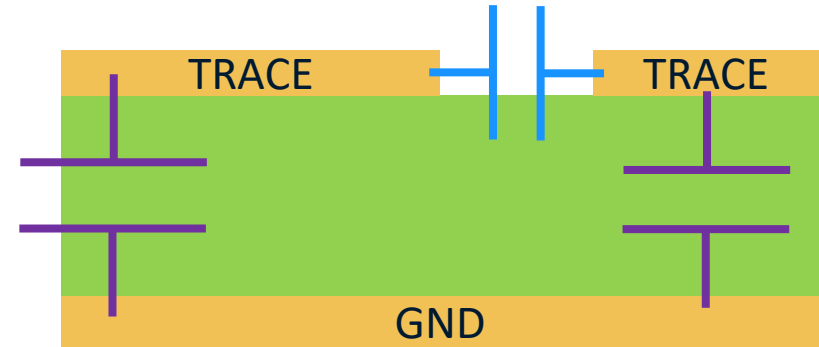
Parasitics

Parasitic Capacitances

- Anywhere in your circuit where you have two signals at different voltage levels separated by a dielectric
- Mostly applies to signals in parallel with one another
- Affects behavior of trace (brings down its impedance)

Reduction Strategies

- Increase capacitance between traces and reference (GND)
 - Use different substrate thickness and/or substrate with different D_k
 - Just add a capacitor (decoupling capacitors)

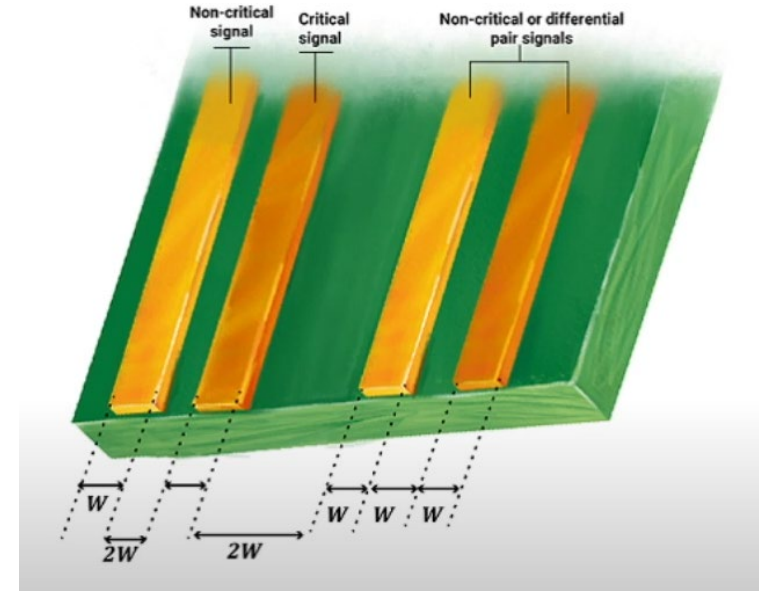


Parasitics

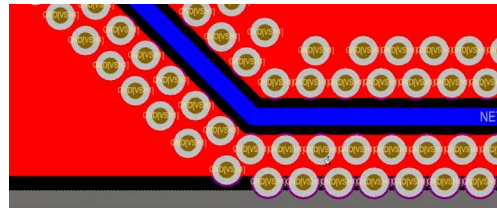
Parasitic Capacitances

Reduction Strategies (cont.)

- Reduce widths of high impedance traces
- Increase spacing between traces
- Avoid routing unrelated or unpaired traces in parallel
- Create faraday shields between traces (E.g. via stitching/shielding)



<https://www.youtube.com/watch?v=sFT2TXIDdxA>



<https://www.youtube.com/watch?v=L9cw-8e--7Q>

Parasitics

Parasitic Inductances

- ⦿ Most prominent in locations where a signal changes direction (coils, vias, trace turns, etc.)
- ⦿ Mostly applies to signals in series with one another

Reduction Strategies

- ⦿ Reduce the equivalent loop area covered by traces
 - Place supply and ground planes/signals as close to critical traces as possible

EMI (Electromagnetic Interference) Reduction

- Have a ground/reference plane(s) in your design
 - If you have multiple ground planes, make sure they're well connected in a manner that minimizes voids and loops
- Keep signals operating at different frequencies separate
- Run signals over a reference plane (such as GND) whenever possible



EMI (Electromagnetic Interference) Reduction

Do traces absolutely need to be routed at 45° angles?

Probably not, unless you're working with ultra-high frequency microwave designs

- In the past, sharp corners caused acid traps during the etching process that caused acid to linger and corrode away more copper
- Poor manufacturing processes also made it more likely that 90° traces would separate from the substrate due to things like heat cycling

EDA software still uses 45° traces by default, so just stick with it

Voiding

- Separate signals need to have a gap in the copper between them so that they aren't electrically connected
- Placing components too close together can create large gaps in the copper pour
 - Lengthens return current paths
 - Increases spread of electric field
- Be aware of this and try not to place vias and through-hole components too close together



<https://www.youtube.com/watch?v=WPT96w3eLAM>

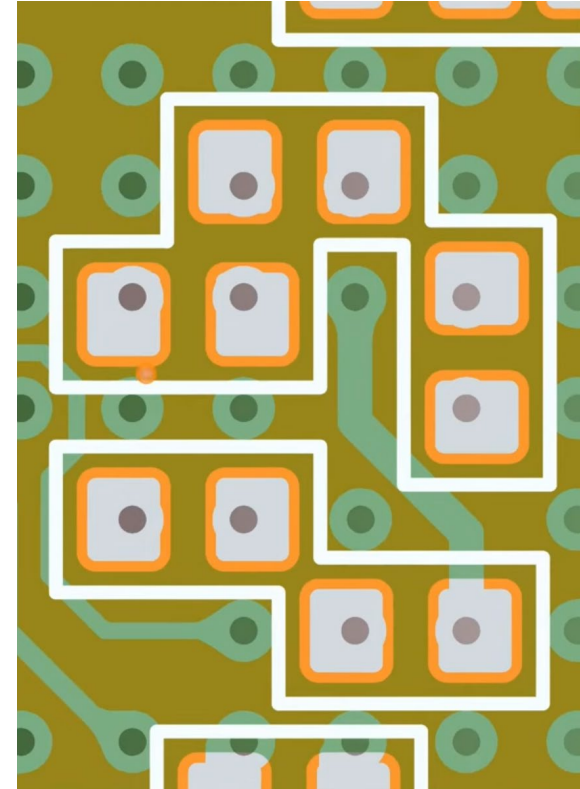
Tombstoning and Via Solder Wicking

It is possible to place vias inside the pads of SMDs

○ Pros

- Saves space
- Reduces parasitic inductance

BUT we want to avoid doing this unless absolutely required, because...



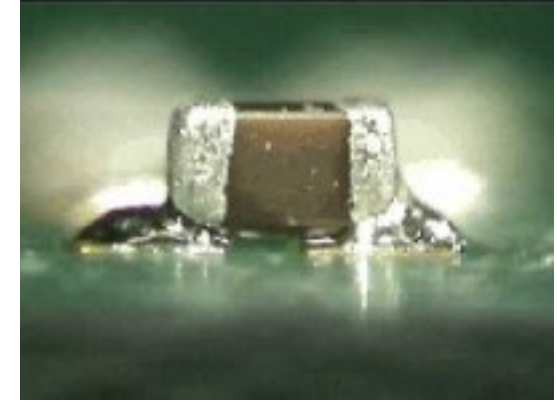
<https://www.youtube.com/watch?v=-8qMEb54os8>

Tombstoning and Via Solder Wicking

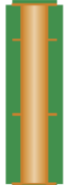
It is possible to place vias inside the pads of SMDs

Cons

- Solder can get wicked into the vias during soldering, creating unreliable connections due to the reduction in solder, and/or tombstoning
- The above can be avoided by having fab fill the holes and cover them with a metalized surface, but this is expensive



An SMD tombstoning during the reflow soldering process

	Standard	Standard	+\$	+\$	++\$	++\$
Cost	Standard	Standard	+\$	+\$	++\$	++\$
Duration	Standard	Standard	+⌚	+⌚	+⌚⌚	+⌚⌚
Type	Through Via	Tented Via	Blind Via	Buried Via	Stacked Via	Via in Pad
						

<https://www.tempoautomation.com/blog/via-in-pad-guidelines-choosing-the-best-routing-for-your-smcs-tempo-automation/>

Arcing

Electricity can arc from one net to another in high voltage systems

- In space, it is possible for arcing to occur at thresholds below those defined by Paschen's Law (describes how breakdown voltage dependent on gas, pressure, and distance)
- Occurs if electric field is strong enough to cause field electron emission

Reduction Strategies

- Ensure appropriate clearance and creepage distances
- Avoid sharp features in traces
- Coat or pot PCB with insulator



<https://www.youtube.com/watch?v=WM2SpUsrODk>

PCB Lab:

Ford ES&T L1255

09/28/2022 - 5:00 PM

